Δεύτερο Σύνολο Ασκήσεων

Η βιβλιογραφία είναι ενδεικτική, οι ομάδες που ενδιαφέρονται πρέπει να περιμένουν να πάρουν από μένα το κατάλληλο υλικό μετά από e_mail σε makri@ceid.upatras.gr με τρεις επιλογές.

Οι εργασίες είναι χωρισμένες σε ομάδες, και θα πρέπει σε μία πρώτη φάση να μας στείλετε τρεις επιλογές και από αυτές θα σας ανατεθεί μία. Οι επιλογές μπορούν να αναφέρονται σε εργασίες από διαφορετικές ομάδες.

Πιο συγκεκριμένα η δήλωση εργασίας, για το δεύτερο project, στην οποία δηλώνετε ΜΕ ΣΕΙΡΑ ΠΡΟΤΙΜΗΣΗΣ ΤΡΕΙΣ ΕΠΙΛΟΓΕΣ, θα πρέπει να γίνει με e-mail στη διεύθυνση makri@ceid.upatras.gr, από 9.3.2026 μέχρι τις 30.3.2026 Η ανάθεση της δεύτερης εργασίας και η αντίστοιχη διανομή υλικού θα πραγματοποιηθούν από μένα με e_mail (το αργότερο μέσα σε μία εβδομάδα από την λήψη του email σας).). Το παραδοτέο της 2ης γ.ε. θα είναι μία αναφορά (μεγέθους 20-30 σελίδες), στην οποία θα καταγράφονται οι βασικές ιδέες των papers της εργασίας που έχετε επιλέξει (επιπλέον βιβλιογραφικό υλικό από εσάς πάνω στο αντικείμενο των αρχικών papers που θα στείλω, θα προσμετρηθεί θετικά).

ΠΡΩΤΗ ΟΜΑΔΑ ΘΕΜΑΤΩΝ

1. Δεικτοδότηση Συμπιεσμένων Βιολογικών Ακολουθιών

Ενδεικτική Βιβλιογραφία

- Francisco Claude, Antonio Fariña, Miguel A. Martínez-Prieto, Gonzalo Navarro: Compressed q-Gram Indexing for Highly Repetitive Biological Sequences. BIBE 2010: 86-91
- Veli Mäkinen, Gonzalo Navarro, Jouni Sirén, Niko Välimäki: Storage and Retrieval of Individual Genomes. RECOMB 2009: 121-137

2. Βάσεις Δεδομένων Βιολογικών Ακολουθιών

Ενδεικτική Βιβλιογραφία

- H. Berman, K. Henrick, H. Nakamura, J. Markley, The worldwide Protein Data Bank (wwPDB) ensuring a single uniform archive of PDB data, Nucleic Acids Research, 2007.
- T. Kulikova, R. Akhtar, et al. EMBL Nucleotide Sequence Database in 2006, Nucleic Acids Research, 2007.

3. Αλγόριθμοι κατηγοριοποίησης Βιολογικών Ακολουθιών

Ενδεικτική Βιβλιογραφία

- G. Ratch, C.S. Ong, Introduction to Machine Learning for Predictive Sequence Analysis
- C. Leslie, E. Eskin, J. Weston, and W.S. Noble. Mismatch string kernels for discriminative protein classification. Bioinformatics, 20(4), 2003.

4. Παρουσίαση αλγορίθμων για εύρεση επαναλήψεων .

Ενδεικτική Βιβλιογραφία

- Gerth Stψlting Brodal, R.B. Lyngsew, C. Pedersen, J. Stoye, Finding Maximal Pairs with Bounded Gaps. In Journal of Discrete Algorithms, Special Issue of Matching Patterns, volume 1(1), pages 77-104, 2000.

- G.S. Brodal, C.S. Pedersen, Finding Maximal Quasiperiodicities in Strings, In Proc. 11th Annual Symposium on Combinatorial Pattern Matching, volume 1848 of Lecture Notes in Computer Science, pages 397-411. Springer Verlag, Berlin, 2000.

5. Παρουσίαση αλγορίθμων για εξελικτικά δέντρα
Ενδεικτική Βιβλιογραφία

- G.S. Brodal, Rolf Fagerberg, Christian N. S. Pedersen, and Anna Ostlin, The Complexity of Constructing Evolutionary Trees Using Experiments In Proc. 28th International Colloquium on Automata, Languages, and Programming, volume 2076 of Lecture Notes in Computer Science, pages 140-151. Springer Verlag, Berlin, 2001.
- G.S. Brodal, R. Fagerberg, C. S. Pedersen, Computing the Quartet Distance Between Evolutionary Trees in Time O(nlogn), In *Algorithmica*, Special Issue on ISAAC 2001, volume 38(2), pages 377-385, 2003.
- Phylogenetic Tree Construction http://linneus20.ethz.ch:8080/5_6.html
- Chapter 17. Dan Gusfield. Algorithms on Strings, Trees, and Sequences: Computer Science and Computational Biology. Cambridge 1997

6. Παρουσίαση αλγορίθμων για Δυναμικό String Matching
Ενδεικτική Βιβλιογραφία

- P. Ferragina, Incremental Text Indexing: a new Data Structure, In Proc. Od 2nd ESA, LNCS 855, pages 495-507, 1994.
- P. Ferragina and R. Grossi, Optimal On-Line Search and Sublinear Time Update in String Matching, In Proc. of IEEE FOCS, pages 604-612, 1995.
- S. Alstrup, G.S. Brodal, T. Rauhe, Pattern Matching in Dynamic Texts, In Proc. of Symposium on Discrete Algorithms, pages 819-828, 2000.
- A. Amir, M. Farach, Z Galil, R. Giancarlo, K. Park, Dynamic Dictionary Matching, Journal of Computer and System Sciences, 49:208-222, 1994.
- R.M. Idury and A.A. Schafer, Dynamic Dictionary Matching with Failure Functions, TCS, 131:295-310, 1994.

ΔΕΥΤΕΡΗ ΟΜΑΔΑ ΘΕΜΑΤΩΝ (ΘΕΜΑΤΑ ΑΠΟ Κ. ΜΕΓΑΛΟΟΙΚΟΝΟΜΟΥ)

Analysis of Gene Expression Maps:
L. An, H. Ling, Z. Obradovic, D.J. Smith, and V. Megalooikonomou, "Learning pair-wise gene functional similarity by multiplex gene expression maps", BMC Bioinformatics, 13(Suppl 3):S1, 2012.
L. An, H. Xie, M.H. Chin, Z. Obradovic, D.J. Smith and V. Megalooikonomou, «Analysis of multiplex gene expression maps obtained by voxelation», BMC Bioinformatics, Vol. 10 (Suppl 4): S10, pp. 1-15, 2009.

Protein Signatures:
C. S. Carvalho, D. Vlachakis, G. Tsiliki, V. Megalooikonomou, S. Kossida. "Protein signatures using electrostatic molecular surfaces in harmonic space"," PeerJ, 1:e185, http://dx.doi.org/10.7717/peerj.185
D. Vlachakis, C. Feidakis, V. Megalooikonomou, S. Kossida, "A two-dimensional visualization tool for amino acid domain sequences"," Theoretical Biology and Medical Modelling, vol. 10, No.14, 2013.
D. Vlachakis, G. Tsiliki, D. Tsagkrasoulis, C. S. Carvalho, V. Megalooikonomou, S. Kossida, "Speeding up the drug discovery process: structural similarity searches using molecular surfaces", EMBNet journal, Bioinformatics in Action, Vol 18, No. 1, 2012.

Drug Design:
D. Vlachakis, P. Fakourelis, V. Megalooikonomou, C. Makris, S. Kossida, "DrugOn: a fully integrated pharmacophore modeling and structure optimization toolkit", PeerJ 3:e725; DOI 10.7717/peerj.725, 2015.
D. Vlachakis, D. Tsagrasoulis, V. Megalooikonomou, S. Kossida, "Introducing Drugster: a comprehensive and fully integrated drug design, lead and structure optimization toolkit", Bioinformatics 29 (1): 126-128, 2013.

Data Mining in Bioinformatics:
A. Amidi, S. Amidi, D. Vlachakis, V. Megalooikonomou, N. Paragios, E. I. Zacharaki, "EnzyNet: enzyme classification using 3D convolutional neural networks on spatial representation", PeerJ 6:e4750; DOI 10.7717/peerj.4750, 2018.
C. Salis, E. Papakonstantinou, K. Pierouli, A. Mitsis, L. Basdeki, V. Megalooikonomou, D. Vlachakis, M. Hagidimitriou, "A genomic data mining pipeline for 15 species of the genus Olea", EMBNet Journal, DOI: https://doi.org/10.14806/ej.24.0.922, 2019.

ΤΡΙΤΗ ΟΜΑΔΑ ΘΕΜΑΤΩΝ

(Θεματικά) Ένα κεφάλαιο (και σχετικά papers) από το βιβλίο Data Analytics in Bioinformatics: A Machine Learning Perspective Editor(s): Rabinarayan Satpathy, Tanupriya Choudhury, Suneeta Satpathy, Sachi Nandan Mohanty, Xiaobo Zhang, 2021, Wiley (https://onlinelibrary.wiley.com/doi/book/10.1002/9781119785620)

- ✓ Nancy Anurag Parasa, Jaya Vinay Namgiri, Sachi Nandan Mohanty, Jatindra Kumar Dash, Introduction to Unsupervised Learning in Bioinformatics (Pages: 35-49)

- ✓ Vrs Jhalia, Tripti Swarnkar, A Critical Review on the Application of Artificial Neural Network in Bioinformatics (Pages: 51-76)

- ✓ Saurav Roy, Ratula Ray, Satya Ranjan Dash, Mrunmay Kumar Giri, Plant Disease Detection Using Machine Learning Tools With an Overview on Dimensionality Reduction (Pages: 109-144)

- ✓ S. Mahapatra, T. Swarnkar, Gene Selection Using Integrative Analysis of Multi-Level Omics Data: A Systematic Review (Pages: 145-171)

- ✓ Sudhansu Shekhar Patra, Om Praksah Jena, Gaurav Kumar, Sreyashi Pramanik, Chinmaya Misra, Kamakhya Narain Singh, Random Forest Algorithm in Imbalance Genomics Classification (Pages: 173-190)

- ✓ Shubham Raj, Swati Singh, Avinash Kumar, Sobhangi Sarkar, Chittaranjan Pradhan, Feature Selection and Random Forest Classification for Breast Cancer Disease (Pages: 191-210)

- ✓ Swati Sucharita, Barnali Sahu, Tripti Swarnkar, A Comprehensive Study on the Application of Grey Wolf Optimization for Microarray Data (Pages: 211-248)

- ✓ V. Aakash, S. Sridevi, G. Ananthi, S. Rajaram, Forecasting of Novel Corona Virus Disease (Covid-19) Using LSTM and XG Boosting Algorithms (Pages: 293-311)

- ✓ P. Poongodi, E. Udayakumar, K. Srihari, Nandan Mohanty Sachi, An Innovative Machine Learning Approach to Diagnose Cancer at an Early Stage (Pages: 313-337)

ΤΕΤΑΡΤΗ ΟΜΑΔΑ ΘΕΜΑΤΩΝ

(Θεματικά) Ένα κεφάλαιο (και σχετικά papers) από το βιβλίο Handbook of Statistical Bioinformatics, Henry Horng-Shing Lu, Bernhard Schölkopf, Hongyu Zhao Springer, 2011 (https://link.springer.com/book/10.1007/978-3-642-16345-6)

✓ **Sequence Analysis**

Accuracy Assessment of Consensus Sequence from Shotgun Sequencing Lei M. Li

Statistical and Computational Studies on Alternative Splicing Liang Chen

Statistical Learning and Modeling of TF-DNA Binding Bo Jiang, Jun S. Liu

Computational Promoter Prediction in a Vertebrate Genome Michael Q. Zhang

Discovering Influential Variables: A General Computer Intensive Method for Common Genetic Disorders Tian Zheng, Herman Chernoff, Inchi Hu, Iuliana Ionita-Laza, Shaw-Hwa Lo

STORMSeq: A Method for Ranking Regulatory Sequences by Integrating Experimental Datasets with Diverse Computational Predictions Jim C. Huang, Brendan J. Frey

Mixture Tree Construction and Its Applications Grace S. C. Chen, Mingze Li, Michael Rosenberg, Bruce Lindsay

✓ **Expression Data Analysis**

Experimental Designs and ANOVA for Microarray Data Richard E. Kennedy, Xiangqin Cui

The MicroArray Quality Control (MAQC) Project and Cross-Platform Analysis of Microarray Data Zhining Wen, Zhenqiang Su, Jie Liu, Baitang Ning, Lei Guo, Weida Tong et al.

A Survey of Classification Techniques for Microarray Data Analysis Wai-Ki Yip, Samir B. Amin, Cheng Li

Statistical Analysis of Single Nucleotide Polymorphism Microarrays in Cancer Studies Pierre Neuvial, Henrik Bengtsson, Terence P. Speed

Computational Analysis of ChIP-chip Data Hongkai Ji

eQTL Mapping for Functional Classes of Saccharomyces cerevisiae Genes with Multivariate Sparse Partial Least Squares Regression Dongjun Chung, Sündüz Keleş

Statistical Analysis of Time Course Microarray Data Lingyan Ruan, Ming Yuan

- ✓ **Systems Biology**

Kernel Methods in Bioinformatics Karsten M. Borgwardt

Graph Classification Methods in Chemoinformatics Koji Tsuda

Hidden Markov Random Field Models for Network-Based Analysis of Genomic Data Hongzhe Li

Review of Weighted Gene Coexpression Network Analysis Tova Fuller, Peter Langfelder, Angela Presson, Steve Horvath

Liquid Association and Related Ideas in Quantifying Changes in Correlation Ker-Chau Li

Boolean Networks Tung-Hung Chueh, Henry Horng-Shing Lu

Protein Interaction Networks: Protein Domain Interaction and Protein Function Prediction Yanjun Qi, William Stafford Noble

Reverse Engineering of Gene Regulation Networks with an Application to the DREAM4 in silico Network Challenge Hyonho Chun, Jia Kang, Xianghua Zhang, Minghua Deng, Haisu Ma, Hongyu Zhao

Inferring Signaling and Gene Regulatory Network from Genetic and Genomic Information Zhidong Tu, Jun Zhu, Fengzhu Sun

Computational Drug Target Pathway Discovery: A Bayesian Network Approach
Seiya Imoto, Yoshinori Tamada, Hiromitsu Araki, Satoru Miyano

Cancer Systems Biology
Elana J. Fertig, Ludmila V. Danilova, Michael F. Ochs

Comparative Genomics Xuhua Xia

Robust Control of Immune Systems Under Noises: Stochastic Game Approach
Bor-Sen Chen, Chia-Hung Chang, Yung-Jen Chuang

ΠΕΜΠΤΗ ΟΜΑΔΑ ΘΕΜΑΤΩΝ

1. Deep Learning in CRISPR – Topics

gRNA Design & On-Target Efficiency Prediction

Wang, D., Zhang, C., Wang, B., Li, B., Wang, Q., Liu, D., Wang, H., Zhou, Y., Shi, L., Lan, F., Wang, Y. (2019).DeepCRISPR: optimized CRISPR guide RNA design by deep learning. Nature Communications volume 10, Article number: 4284 (2019)

Kim, H. K., Min, S., Song, M., Jung, S., Choi, J. W., Kim, Y., Lee, S., Yoon, S., & Kim, H. H. (2019). DeepSpCas9, a deep learning-based model for predicting SpCas9 activity. Science Advances, 5(9), eaax9249.

Wang, D.; Zhang, C.; Wang, B.; Li, B.; Wang, Q.; Liu, D.; Wang, H.; Zhou, Y.; Shi, L.; Lan, F.; et al. Optimized CRISPR guide RNA design for two high-fidelity Cas9 variants by deep learning. *Nat. Commun.* **2019**, *10*, 4284.**2.**

Off-Target Prediction Using Deep Learning

Lin, J., Wong, K. C. (2019). CRISPR-Net: A recurrent convolutional network for guide RNA off-target prediction. Advanced Sience 2020.

Tsai, S. Q., Nguyen, N. T., Malagon-Lopez, J., Topkar, V. V., Aryee, M. J., & Joung, J. K. (2017). CIRCLE-seq: A highly sensitive in vitro screen for genome-wide CRISPR–Cas9 nuclease off-targets. Nature Methods volume 14, pages607–614 (2017)

Base Editing & Prime Editing Prediction

M Arbab, MW Shen, B Mok, C Wilson, Ż Matuszek, CA Cassa, DR Liu, Determinants of base editing outcomes from target library analysis and machine learning Cell 182 (2), 463-480. e30

Kim N*, Choi S*, Kim S, Song M, Seo JH, Min S, Park J, Cho S-R, Kim HH+. Deep learning models to predict the editing efficiencies and outcomes of diverse base editors. Nat. Biotechnol. 2024. 1. 42(3):484-497

Protein Engineering & Single-Cell CRISPR Analysis

Rives, A., Meier, J., Sercu, T., et al. (2021).
Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences.
Proc Natl Acad Sci U S A . 2021 Apr 13;118(15):e2016239118. doi: 10.1073/pnas.2016239118.

Dixit, A., Parnas, O., Li, B., et al. (2016).
Perturb-Seq: Dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens.
Cell, 167(7), 1853–1866.

2. **Quantum Algorithms in Bioinformatics**

Yidong Zhou et al. Quantum-machine-assisted drug discovery, npj Drug Discovery, 2026, https://www.nature.com/npjdrugdiscov/

Abdullah et al. Performance Evaluation of Quantum Machine Learning Classifiers in Bioinformatics Applications December 2025 DOI: 10.1109/BTS-I2C67944.2025.11399503

Gundala Pallavi, Rangarajan Prasanna Kumar *Quantum natural language processing and its applications in bioin formatics: a comprehensive review of methodologies, concepts, and future directions, Front. Comput. Sci., 18 February 2025